

Estimation and confidence intervals

Q) Build 99% confidence interval using sample standard deviation

→ given data

- Sample durability values (in millions of character)

1.13, 1.55, 1.43, 0.92, 1.25, 1.36, 1.32, 0.95, 1.07, 1.48
1.20, 1.33, 1.18, 1.22, 1.29

- Sample size :- $n = 15$

- Confidence level :- 99%

→ Since the population standard deviation is not known and the sample size is small ($n < 30$), the t-distribution is used

→ Step 1 :- Calculate sample Mean

$$\bar{x} = \frac{\sum x}{n} = \frac{18.58}{15}$$
$$= 1.2387$$

→ Step 2 :- Calculate sample standard deviation

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

$$= 0.209$$

Step 3 :- Determine critical t-value

Confidence level = 99%

Therefore $\alpha = 0.01$

Degrees of free dom :-

$$\begin{aligned}df &= n-1 \\&= 15-1 \\&= 14\end{aligned}$$

from the t-table

$$t_{0.005, 14} = 2.977$$

Step 4:- calculate the standard error

$$\begin{aligned}SE &= \frac{s}{\sqrt{n}} \\&= \frac{0.204}{\sqrt{15}} \\&= 0.054\end{aligned}$$

Step 5:- calculate margin of error

$$\begin{aligned}ME &= t \times SE \\&= 2.977 \times 0.054 \\&= 0.161\end{aligned}$$

Step 7:- construct confidence ~~level~~ interval

$$\text{lower limit} :- 1.2387 - 0.161 = 1.078$$

$$\text{upper limit} :- 1.2387 + 0.161 = 1.400$$

$\therefore (1.078, 1.400)$ million characters

Interpretation

We are 99% confident that the true mean durability of the print-head lies between 1.078 and 1.400 million characters before failure.

b) 99.1. Confidence interval using known Population Standard Deviation

→ Given $\sigma = 0.2$

Since the population standard deviation is known the z-distribution is used

The confidence interval formula

$$\bar{x} \pm z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

→ step 1:- Sample mean

$$\bar{x} = 1.2387$$

→ step 2:- Determine critical z-value

for a 99.1. confidence level

$$z_{0.005} = 2.576$$

step 3:- calculate standard Error

$$SE = \frac{0.2}{\sqrt{15}}$$

$$= 0.0516$$

→ step 4:- calculate margin of error

$$ME = 2.576 \times 0.0516$$

$$= 0.133$$

→ step 5:- construct confidence Interval

$$\text{lower limit} = 1.2387 - 0.133 = 1.106$$

$$\text{upper limit} = 1.2387 + 0.133 = 1.372$$

∴ (1.106, 1.372) million characters

Interpretation

we are 99.1% confident that the true mean durability of the print-heads lies between 1.106 and 1.372 million characters before failure.

→ How Confidence Intervals change - when population standard deviation is unknown vs known

↳ The difference comes from how certain we are about the variability of the population

↳ When the population standard deviation is known, the z -distribution is used to construct the CI. Since the variability of the population is known exactly, the resulting confidence interval is relatively narrow.

↳ when the population standard deviation is unknown, it must be estimated using sample standard deviation.

This introduces additional uncertainty and therefore t -distribution is used. The t -distribution has heavier tails and produces larger critical values resulting in wider confidence intervals.